DUNMAN SECONDARY SCHOOL

CANDIDATE NAME			
CLASS		INDEX NUMBER	

PRELIMINARY EXAMINATION 2024
SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A — 4045/02

Paper 2

5 August 2024

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Calculator Model

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.
Omission of essential working will result in loss of marks.
The total number of marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator or 3.142.

Total Marks

Mathematical Formulae

Compound interest

$$\text{Total amount} = P\left(1+\frac{r}{100}\right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi rl$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2}ab\sin C$$

$$\text{Arc length} = r\theta\text{, where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2}r^2\theta\text{, where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

Section A (62 marks)

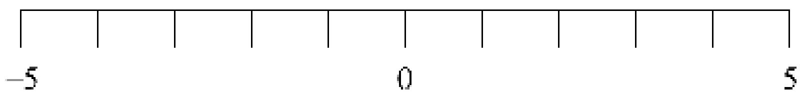
Answer **all** the questions in this section.

1 **(a)** Work out $\dfrac{(-8)^2 - \sqrt{6-7\times3\times(-2)}}{4}$.

Leave your answer correct to 2 decimal places.

Answer .... [1]

(b) Mark a (x) on the number line at the smallest prime number.



[1]

(c) Arrange the following weights in order, starting with the largest.

9248 mg   724 g   0.38 kg

Answer .... , ... , ... [2]

Largest

2 **(a)** Write $\frac{3\times 3^{-6}}{9^2}$ as a single power of 3.

Answer .... [2]

(b) Simplify $\left(\frac{125}{a^9}\right)^{-\frac{2}{3}}$.

Answer .... [2]

3 Amy carried out a survey with her classmates on the number of hours they spent studying during the weekend.

Number of hours	0	1	2	3	4	5
Frequency	3	5	12	p	6	2

(a) If the mean is 2.4 hours, find the value of p.

Answer $p =$ ……………………… [3]

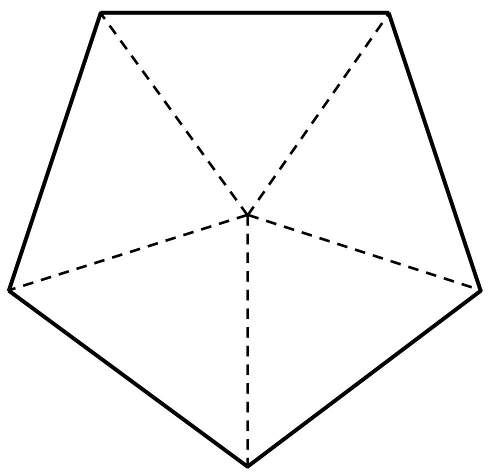
(b) A classmate is selected at random.
Using the value of p from part **(a)**, what is the probability that the selected student studied **more than** 3 hours during the weekend.

Answer ………………………… [1]

4 **(a)** Find the interior angle of a regular pentagon.

Answer ……………………… ° [1]

(b) A regular pentagon has a side of 8 cm.



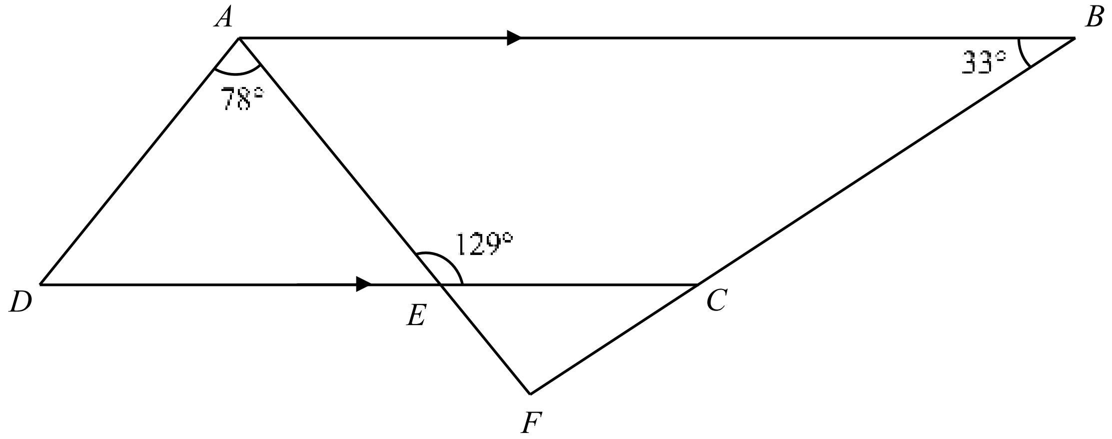
Calculate the area of the pentagon.

Answer …………………… cm^2 [3]

(c) The base of a prism is a regular pentagon with side of 8 cm.
The volume of the prism is 3528.5 cm^3.
Calculate the height of the prism.
Leave your answer correct to 3 significant figures.

Answer …………………… cm [2]

5



In the diagram, *AEF*, *DEC* and *BCF* are straight lines.
AB is parallel to *CD*.
Angle $DAE = 78°$, angle $AEC = 129°$ and angle $ABF = 33°$.

(a) Find

(i) angle *FAB*,

Answer Angle *FAB* = …………… ° [1]

(ii) angle *AFB*,

Answer Angle *AFB* = …………… ° [1]

(iii) angle *FCE*,

Answer Angle *FCE* = …………… ° [1]

(b) Explain why triangle *ADE* is isosceles.
Show all your working.

Answer

[2]

6 **(a)** Rearrange this formula to make x the subject.

$$y = \frac{x-b}{ax}$$

Answer $x =$ ……………………… [3]

(b) Factorise $y^2(y+7)-16(y+7)$ completely.

Answer ……………………… [2]

(c) A mobile phone shop sells screen protectors at $\$x$ each.
The cost of a phone casing is \$4.50 more than a screen protector.
Mark bought 2 protector and 5 casing from the shop and paid \$75.
Find the cost of one phone casing.

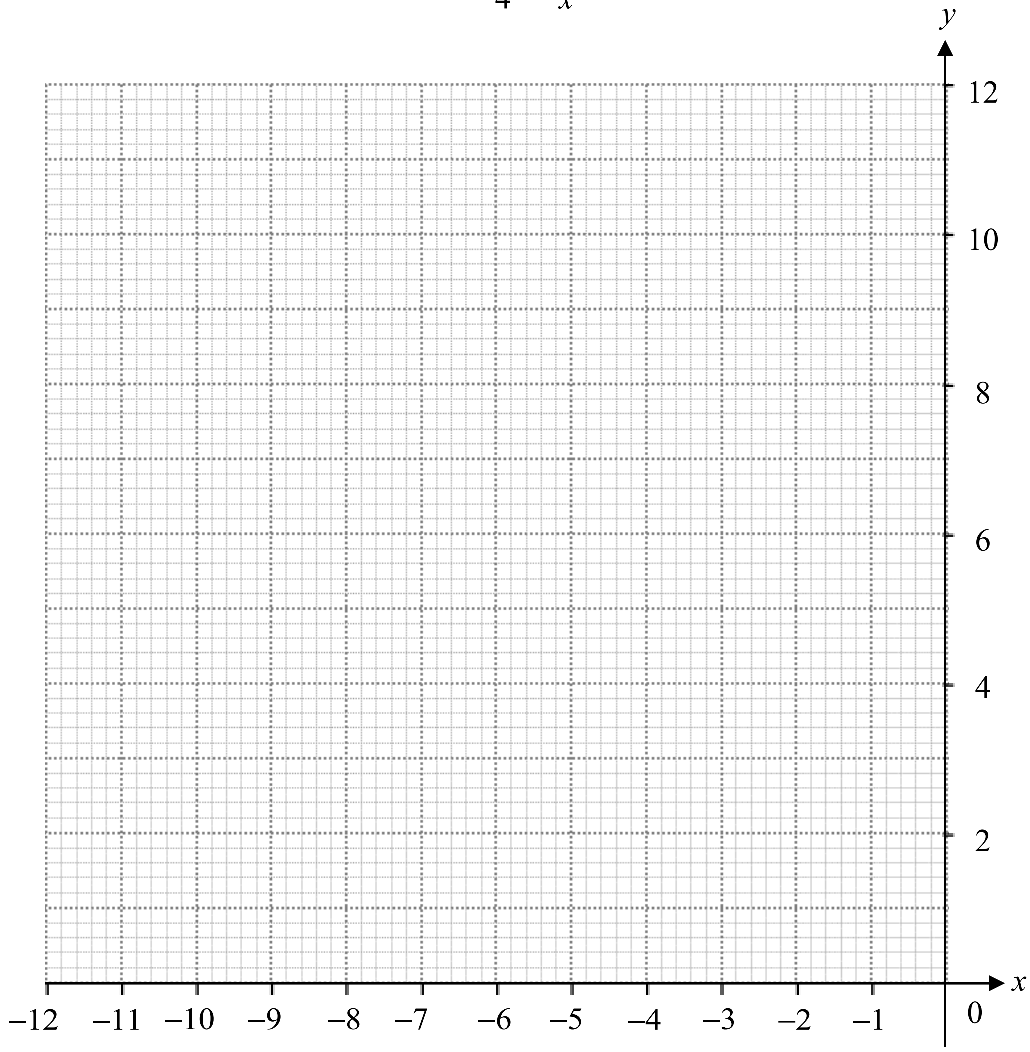
Answer \$ ……………………… [3]

7 **(a)** Complete the table of values for $y = -\frac{3x}{4} - \frac{12}{x}$.

x	−12	−10	−8	−6	−4	−2.4	−2	−1.2
y		8.7	7.5	6.5	6	6.8	7.5	10.9

[1]

(b) On the grid, plot the graph of $y = -\frac{3x}{4} - \frac{12}{x}$ for $-12 \le x \le -1.2$.



[2]

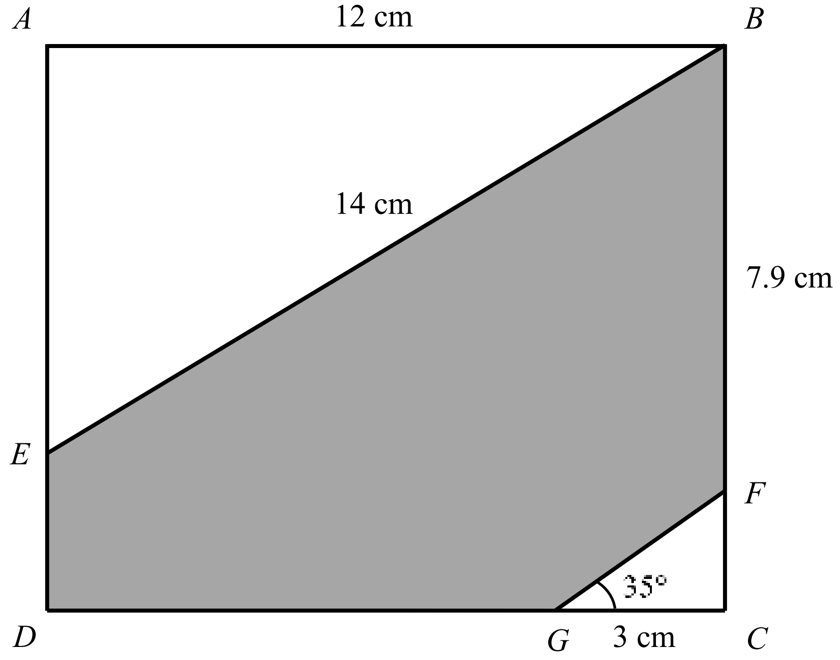
(c) Draw a suitable tangent on the graph such that the gradient of the tangent is 0.
State the value of x.

Answer $x =$ …………………… [2]

(d) Using your graph, solve the equation $-\frac{3x}{4}-\frac{12}{x}=7$.

Answer $x=$ …………… and …………… [3]

8



In the diagram, $ABCD$ is a rectangle.
$AB = 12$ cm, $BF = 7.9$ cm, $CG = 3$ cm and $BE = 14$ cm.
Angle $CGF = 35°$.

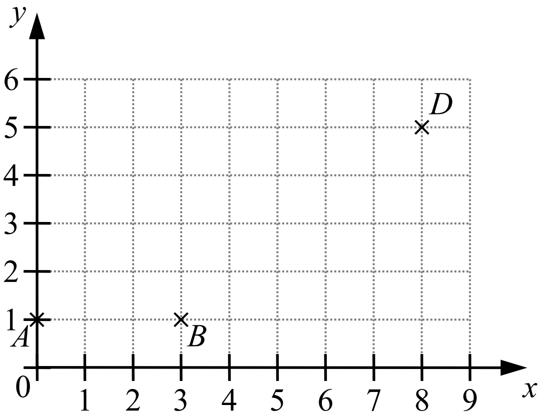
(a) Find the length of AE.

Answer …………………… cm [2]

(b) Find the area of the shaded region.

Answer …………………… cm^2 [4]

9



A, *B* and *D* are the points (0, 1), (3, 1) and (8, 5) respectively.
BCD is a triangle where *CD* is parallel to the *y* – axis and length of *CD* is 3 units.

(a) Find the coordinate of *C*.

Answer (………… , …………) [1]

(b) Find the equation of

(i) *AB*,

Answer ……………………… [1]

(ii) *BD*.

Answer ……………………… [3]

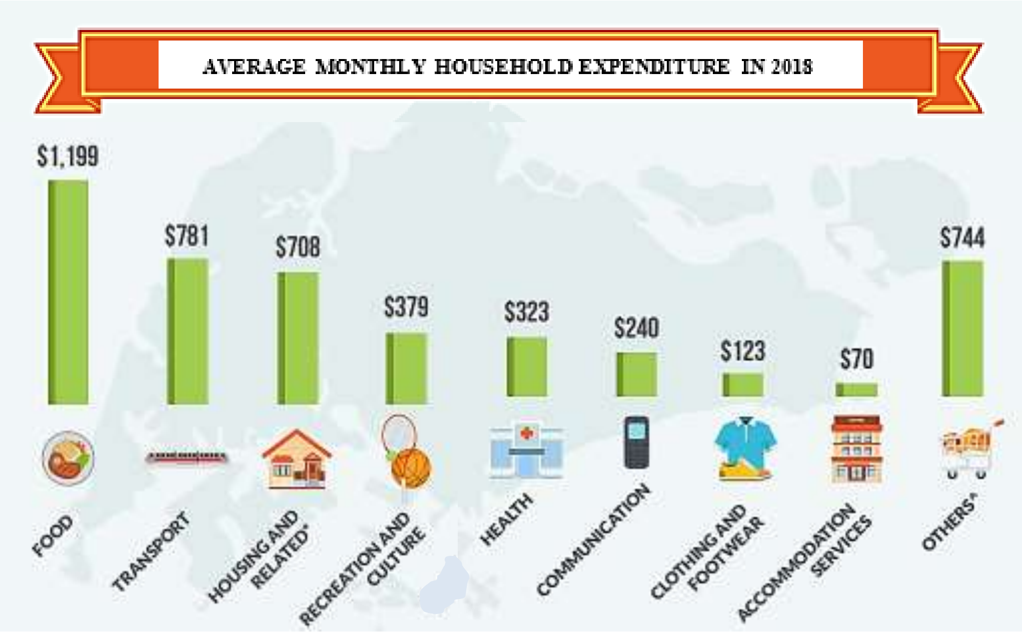
(c) Find the area of *ABCD*.

Answer …………………… unit^2 [2]

(d) Find the length of *AD*.

Answer AD =……………… unit [2]

10 The bar graph shows Singapore's **average monthly household expenditure** in 2018.



(a) Express the food expenditure as a percentage of the **average monthly household expenditure**.

Answer ……………… % [2]

The table shows the estimated yearly average cost of studying in a University in 2 countries, Korea and Sweden. The estimated yearly average cost is made up of school fee and living expenses.

Country	Currency	School Fee	Living Expenses
Korea	Won (₩)	₩17810000	₩17673000
Sweden	Korna (kr)	182200 kr	129000 kr

(b) Given that S$1 = ₩1024.15, calculate the estimated yearly average cost of studying in a university in Korea in S$.

Answer S$……………… [2]

(c) Mr Mah wishes to send his child to either Korea or Sweden to study in a university for a year.

His household income is shown in the table.

Income Type	Frequency	Amount ($)
Salary	Monthly	6950
Performance Bonus	Yearly	1.25 of salary

Based on Singapore's average monthly household expenditure in 2018 and given that 1 kr = S$0.1247, which country, Korea or Sweden, can Mr Mah afford to send his child for studying in a university.
Show working to support you answer.
Answer

Mr Mah can afford to send his child to ……………………………………………………

……………………………………………………………………………………………… [4]

Section B (8 marks)

Answer **one** question from this section. Each question carries 8 marks.

11 **(a)** The table summarises the marks of the 180 students of School A in a Mathematics examination.

Mark (x)	$0 < x \leq 10$	$10 < x \leq 20$	$20 < x \leq 30$	$30 < x \leq 40$	$40 < x \leq 50$	$50 < x \leq 60$
Frequency	9	28	41	53	34	15

(i) Calculate an estimate of the mean and standard deviation of these marks

Answer Mean = ………………

Standard deviation = ……………… [3]

The students in School B sat for the same examination.
For School B, the mean marks of the students was 40.2 and the standard deviation was 12.9.

(iii) Which school's Mathematics result has a wider variation in their marks. Explain your answer.

Answer ………………………………………………………………………………

………………………………………………………………………………

……………………………………………………………………………… [2]

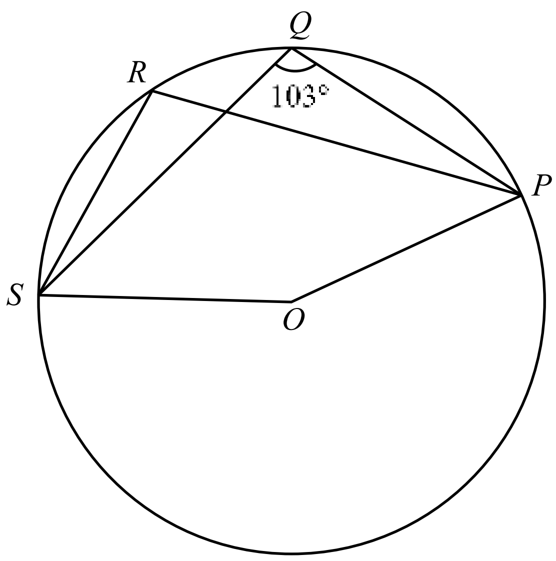
(b) A box contains 4 red balls and 3 green balls. One ball is drawn at random.

If it is red, it is replaced back to the box.
However, if it is green, it is not replaced back to the box.
A second ball is then drawn from the box.

Calculate the probability that the balls are of different colours.

Answer …………………… [3]

12



(a) P, Q, R and S are points on the circle centre O.

Angle PQS is $103°$.

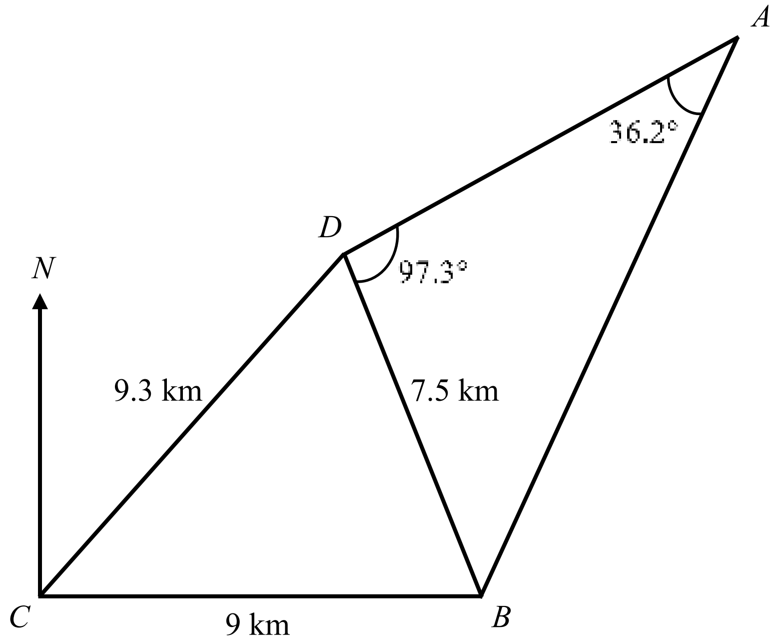
(i) Find angle SRP.

Answer Angle SRP = …………… ° [1]

(ii) Find angle SOP.

Answer Angle SOP = …………… ° [2]

(b) In the diagram below, A, B, C and D are four points on level ground.
Point B is due east of point C.
$BC = 9$ km, $BD = 7.5$ km, $CD = 9.3$ km, angle $ADB = 97.3°$ and angle $BAD = 36.2°$.



Calculate

(i) the length of AB,

Answer …………………… km [2]

(ii) angle DBC,

Answer …………………… ° [2]

(iii) the bearing of D from B.

Answer …………………… ° [1]

END OF PAPER

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